NOTES ON KÄHLER GEOMETRY

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Abstract. This note was taken for the class on Calabi-Yau Theorem in Complex Geometry given by Professor Ben Weinkove (Northwestern University) in the summer school in the USTC, July 2007.

This note includes the following topics on Kähler Geometry and the main reference is [1].

- Complex Manifolds.
- Tangent Spaces, Vector Fields, Cotangent Spaces, Covector Fields.
- Metrics, Covariant Differentiation, Curvature, Differential Forms.
- Holomorphic Line Bundles.
- Coordinate Systems, Integration.
- Curvature of Hermitian Metrics, Chern Class.
- Yau's Theorem(Omit proof).

1. Complex Manifolds

Complex manifold of dimension n:

- Real manifold of dimension $2n$, locally homeormorphic to $\mathbb{R}^{2n} \cong \mathbb{C}^n$.
- Coordinate transformations are holomorphic.

Precisely:

M is a complex manifold of complex dimension n if M can be covered by charts (U, φ) where U is an open subset of M and $\varphi : U \longrightarrow \mathbb{C}^n$ is a homeomorphism onto an open subset $\varphi(U)$ of $\mathbb{C}$, with the following property: assume $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β) are different charts, $U_\alpha \cap U_\beta \neq \emptyset$, then the transition maps:

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \quad \varphi_\alpha(U_\alpha \cap U_\beta) \longrightarrow \varphi_\beta(U_\alpha \cap U_\beta)$$

and

$$\varphi_\alpha \circ \varphi_\beta^{-1} : \quad \varphi_\beta(U_\alpha \cap U_\beta) \longrightarrow \varphi_\alpha(U_\alpha \cap U_\beta)$$

are holomorphic.

We write $\varphi = z = (z^1, \cdots, z^n)$ is called complex coordinates. The real coordinates $(x^1, \cdots, x^n, y^1, \cdots, y^n)$ defined by the usual formula

$$z^i = x^i + \sqrt{-1} y^i.$$

Example 1. $\mathbb{C}P^n$*: complex projective space.*

$\mathbb{C}P^n := \mathbb{C}^{n+1} \setminus \{0\}/\sim$*, where* $\sim$: $(z_0, z_1, \cdots, z_n) \sim (z_0', z_1', \cdots, z_n')$ *if* $z_i = \lambda z_i', \lambda \neq 0$.

Write elements of $\mathbb{C}P^n$ on $[z_0, z_1, \cdots, z_n]$(The z_i are called homogeneous coordinates). Take cover

$$U_\alpha = \{[z_0, z_1, \cdots, z_n] \in \mathbb{C}P^n \mid z_\alpha \neq 0\}.$$

Define

$$\begin{aligned} \varphi_\alpha : \quad U_\alpha &\longrightarrow \mathbb{C}^n \\ [z_0, z_1, \cdots, z_n] &\longmapsto \left(\frac{z_0}{z_\alpha}, \cdots, \frac{z_{\alpha-1}}{z_\alpha}, \frac{z_{\alpha+1}}{z_\alpha}, \cdots, \frac{z_n}{z_\alpha}\right) \end{aligned}$$

Check $\mathbb{C}P^n$ is complex manifold:

$$\begin{tikzcd} (w^1, w^2, \cdots, w^n) \arrow[r, "\varphi_\alpha^{-1}"] & {[w^1, w^2, \cdots, w^{\alpha-1}, w^\alpha, 1, w^{\alpha+1}, \cdots, w^n]} \arrow[d, "\varphi_\beta"] \\ & \left(\frac{w^1}{w^\beta}, \frac{w^2}{w^\beta}, \cdots, \frac{w^{\beta-1}}{w^\beta}, \frac{w^{\beta+1}}{w^\beta}, \cdots, \frac{w^\alpha}{w^\beta}, \frac{1}{w^\beta}, \frac{w^{\alpha+1}}{w^\beta}, \cdots, \frac{w^n}{w^\beta}\right) \end{tikzcd}$$

Since $w^\beta \neq 0$, then $\varphi_\beta \circ \varphi_\alpha^{-1}$ is holomorphic map.

2. Tangent Spaces

Let M be a real smooth manifold of dimension $2n$. Fix a point $p \in M$, then $T_pM = \left\{\frac{\partial}{\partial x^1}, \cdots, \frac{\partial}{\partial x^{2n}}\right\}$ span as a real vector space, i.e., **tangent space** of M. Consider complexification, $T_p^{\mathbb{C}}M = T_pM \otimes \mathbb{C} = \left\{\frac{\partial}{\partial x^1}, \cdots, \frac{\partial}{\partial x^{2n}}\right\}_{\mathbb{C}}$ span as a complex vector space, i.e., **tangent space** of M.

We define operators $\frac{\partial}{\partial z^i}$and $\frac{\partial}{\partial \bar{z}^i}$ on $\mathbb{C}^n$ by

$$\frac{\partial}{\partial z^i} = \frac{1}{2}\left(\frac{\partial}{\partial x^i} - \sqrt{-1}\frac{\partial}{\partial x^{n+i}}\right), \quad \frac{\partial}{\partial \bar{z}^i} = \frac{1}{2}\left(\frac{\partial}{\partial x^i} + \sqrt{-1}\frac{\partial}{\partial x^{n+i}}\right).$$

So we can wirte

$$T_p^{\mathbb{C}}M = \left\{\frac{\partial}{\partial z^i}, \cdots, \frac{\partial}{\partial z^n}, \frac{\partial}{\partial \bar{z}^1}, \cdots, \frac{\partial}{\partial \bar{z}^n}\right\}.$$

As the notation suggests, we have for all $i, j = 1, \cdots, n$,

$$\frac{\partial}{\partial z^i}(z^j) = \delta_{ij}, \quad \frac{\partial}{\partial z^i}(\bar{z}^j) = 0, \quad \frac{\partial}{\partial \bar{z}^i}(z^j) = 0, \quad \frac{\partial}{\partial \bar{z}^i}(\bar{z}^j) = \delta_{ij},$$

where δ_{ij} is the Kronecker delta symbol. In particular, a smooth function f on $\mathbb{C}^n$ is holomorphic if and only if

$$\frac{\partial f}{\partial \bar{z}^i} = 0, \quad \text{for } i = 1, \cdots, n.$$

We write two charts (U, z), $(\tilde{U}, \tilde{z})$ with $U \cap \tilde{U} \neq \emptyset$ where $z = (z^1, \cdots, z^n)$ and $\tilde{z} = (\tilde{z}^1, \cdots, \tilde{z}^n)$.

Hence the condition that the transition maps be holomorphic can be written as

$$\frac{\partial \tilde{z}^i}{\partial \bar{z}^j} = 0, \quad \frac{\partial z^i}{\partial \bar{\tilde{z}}^j} = 0, \quad \text{for all } i, j = 1, \cdots, n.$$

Here we are writing $\tilde{z}$ for the map $\tilde{z} \circ z^{-1}$ and z for the map $z \circ \tilde{z}^{-1}$.

A smooth function f on M is **holomorphic** if for each coordinate chart (U, z) we have

$$\boxed{\frac{\partial f}{\partial \bar{z}^i} = 0, \quad \text{for } i = 1, \cdots, n,}$$

on U. Here we are writing f for $f \circ z^{-1}$. It is well defined. Indeed, suppose that $\tilde{U}$ is an overlapping coordinate chart and compute using the chain rule on $U \cap \tilde{U}$,

$$\frac{\partial f}{\partial \bar{\tilde{z}}^j} = \frac{\partial f}{\partial \bar{z}^k} \frac{\partial \bar{z}^k}{\partial \bar{\tilde{z}}^j} + \frac{\partial f}{\partial z^k} \frac{\partial z^k}{\partial \bar{\tilde{z}}^j} = 0,$$

as required.

We write $T_p^{\mathbb{C}} M = T_p^{0,1} M \oplus T_p^{0,1} M$, where $T_p^{1,0} M$ is given by the span of the $\frac{\partial}{\partial z^i}$ and $T_p^{0,1} M$ by the $\frac{\partial}{\partial \bar{z}^i}$. This decomposition of $T_p^{\mathbb{C}} M$ is independent of choice of complex coordinate chart. Indeed,

$$\frac{\partial}{\partial z^i} = \frac{\partial \tilde{z}^k}{\partial z^i} \frac{\partial}{\partial \tilde{z}^k} + \frac{\partial \bar{\tilde{z}}^k}{\partial z^i} \frac{\partial}{\partial \bar{\tilde{z}}^k} = \frac{\partial \tilde{z}^k}{\partial z^i} \frac{\partial}{\partial \tilde{z}^k} \in \operatorname{span}\left\{ \frac{\partial}{\partial \tilde{z}^1}, \cdots, \frac{\partial}{\partial \tilde{z}^n} \right\}$$

and similarly for $\frac{\partial}{\partial \bar{z}^i}$.

3. Vector Fields

We define a $T^{1,0}$ **vector field** on M to be a smooth complex-valued vector field X on M with the property that $X_p \in T_p^{1,0} M$ for all $p \in M$. We write X locally as

$$X = \sum_i X^i \frac{\partial}{\partial z^i},$$

where the $X^i : U \longrightarrow \mathbb{C}$ are smooth functions which satisfy the following transformation rule. If we write $\tilde{X}^i$ for the corresponding functions on $\tilde{U}$ then

$$X^i = \sum_j \tilde{X}^j \frac{\partial z^i}{\partial \tilde{z}^j}, \quad \text{on } U \cap \tilde{U}.$$

Any collection of functions $X^i : U \longrightarrow \mathbb{C}$ defined on each chart in a cover of M, which satisfy the above transformation rule, determine a globally defined vector field X. Indeed one can check that

$$\sum_i X^i \frac{\partial}{\partial z^i} = \sum_j \tilde{X}^j \frac{\partial}{\partial \tilde{z}^j}, \quad \text{on } U \cap \tilde{U}.$$

A $T^{1,0}$ vector field $X = X^i \frac{\partial}{\partial z^i}$ is holomorphic if

$$\frac{\partial X^i}{\partial \bar{z}^j} = 0, \quad \text{for all } i, j = 1, \cdots, n.$$

This condition is well-defined. Indeed

$$\begin{aligned}
\frac{\partial X^i}{\partial \bar{z}^j} &= \frac{\partial}{\partial \bar{z}^j}\left(\tilde{X}^k \frac{\partial z^i}{\partial \tilde{z}^k}\right) \\
&= \frac{\partial \tilde{X}^k}{\partial \bar{z}^j}\frac{\partial z^i}{\partial \tilde{z}^k} + \tilde{X}^k \frac{\partial}{\partial \bar{z}^j}\left(\frac{\partial z^i}{\partial \tilde{z}^k}\right) \\
&= \frac{\partial \bar{\tilde{z}}^l}{\partial \bar{z}^j}\left(\frac{\partial}{\partial \bar{\tilde{z}}^l}\tilde{X}^k\right)\frac{\partial z^i}{\partial \tilde{z}^k}
\end{aligned}$$

and

$$\frac{\partial X^i}{\partial \bar{z}^j} = 0 \Longleftrightarrow \frac{\partial \tilde{X}^k}{\partial \bar{\tilde{z}}^l} = 0.$$

We also define a $T^{0,1}$ vector field on M to be a smooth complex-valued field Y on M with the property that $Y_p \in T_p^{0,1}M$ for all $p \in M$. We write Y locally as

$$Y = \sum_j Y^{\bar{j}} \frac{\partial}{\partial \bar{z}^j},$$

where the $Y^{\bar{j}} : U \longrightarrow \mathbb{C}$ are smooth functions which satisfy the following transformation rule. If we write $\tilde{Y}^{\bar{j}}$ for the corresponding functions on $\tilde{U}$ then

$$Y^{\bar{j}} = \sum_l \tilde{Y}^{\bar{l}} \frac{\partial \bar{z}^j}{\partial \bar{\tilde{z}}^l} \quad \text{on } U \cap \tilde{U}.$$

Example 2. $\mathbb{C}P^1$.
Computing the holomorphic vector field.
Set $U_0 = \{[z_0, z_1 \mid z_0 \neq 0]\}$ *and* $U_1 = \{[z_0, z_1] \mid z_1 \neq 0\}$. *Consider the homeomorphism map*

$$\begin{aligned}
\varphi_0 : \quad \mathbb{C}P^1 &\longrightarrow \mathbb{C} \\
[z_0, z_1] &\longmapsto \frac{z_1}{z_0} = z_{(0)}
\end{aligned}$$

and

$$\begin{aligned}
\varphi_1 : \quad \mathbb{C}P^1 &\longrightarrow \mathbb{C} \\
[z_0, z_1] &\longmapsto \frac{z_0}{z_1} = z_{(1)}.
\end{aligned}$$

We have $z_{(0)} = \frac{1}{z_{(1)}}$ *on* $U_0 \cap U_1$.

Consider $\{V_{(0)}, V_{(1)}\}$ *a holomorphic field. Set* $V_{(0)} = \sum_{i=0}^{\infty} a_i z_{(0)}^i$, *where* $a_i \in \mathbb{C}$ *and*

$$V_{(1)} = V_{(0)} \frac{\partial z_{(1)}}{\partial z_{(0)}} = -V_{(0)} \frac{1}{z_{(0)}^2} = -\sum_{i=0}^{\infty} a_i z_{(0)}^{i-2} = -\sum_{i=0}^{\infty} a_i z_{(1)}^{2-i}.$$

Must have $a_3, a_4, \cdots = 0$, *then*

$$V_{(0)} = a_0 + a_1 z_{(0)} + a_2 z_{(0)}^2, \quad V_{(1)} = -a_0 z_{(1)}^2 - a_1 z_{(1)} - a_2.$$

4. Cotangent Spaces

Let M be a real smooth manifold of dimension $2n$. Fix a point $p \in M$, then $T_pM^* = \{dx^1, \cdots, dx^{2n}\}$, i.e., **cotangent space** of M.
Consider complexification, $T_p^{\mathbb{C}}M^* = T_pM^* \otimes \mathbb{C} = \{dx^1, \cdots, dx^{2n}\}_{\mathbb{C}}$, i.e., **cotangent space** of M.
Recall:

$$dx^i \left(\frac{\partial}{\partial x^j} \right) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j. \end{cases}$$

We define operators dz^i and $d\bar{z}^i$ by

$$dz^i = dx^i + \sqrt{-1} dx^{n+i}, \quad d\bar{z}^i = dx^i - \sqrt{-1} dx^{n+i}.$$

Notice that dz^i and $d\bar{z}^i$ are dual to $\frac{\partial}{\partial z^i}$ and $\frac{\partial}{\partial \bar{z}^i}$ respectively.
So we can write

$$T_p^{\mathbb{C}}M^* = \{dz^1, \cdots, dz^n, d\bar{z}^1, \cdots, d\bar{z}^n\}.$$

We write $T_p^{\mathbb{C}}M^* = T_p^{0,1}M^* \oplus T_p^{0,1}M^*$, where $T_p^{1,0}M^*$ is given by the span of the dz^i and $T_p^{0,1}M^*$ by the $d\bar{z}^i$.

5. Covector Fields

We define a $T^{*1,0}$ **covector field** on M with the property that $a \in T_p^{*1,0}M$ for all $p \in M$. We write a locally as

$$a = \sum_i a_i dz^i$$

where the $a_i : U \longrightarrow \mathbb{C}$ are smooth functions which satisfy the following transformation rule. If we write $\tilde{a}_i$ for the correspomding functions on $\tilde{U}$ then

$$a_i = \sum_k \tilde{a}_k \frac{\partial \tilde{z}^k}{\partial z^i}, \quad \text{on } U \cap \tilde{U}.$$

Any collection of dunctions $a^i : U \longrightarrow \mathbb{C}$ defined on each chart in a cover of M, which satisfy the above transformation rule, determine a globally defined covector field a.

We also define a $T^{*0,1}$ covector field on M with the property that $b \in T_p^{*0,1}M$ for all $p \in M$. We write b locally as

$$b = \sum_j b_{\bar{j}} d\bar{z}^j$$

where the $b_{\bar{j}} : \tilde{U} \longrightarrow \mathbb{C}$ are smooth functions which functions which satisfy the following transformation rule. If we write $\tilde{b}_{\bar{l}}$ for the corresponding functions on $\tilde{U}$ then

$$b_{\bar{j}} = \sum_l \tilde{b}_{\bar{l}} \frac{\partial \bar{\tilde{z}}^l}{\partial z^j}, \quad \text{on } U \cap \tilde{U}.$$

Consider more general tensor fields:

$$T = T_{\bar{k}}^{ij} \frac{\partial}{\partial z^i} \otimes \frac{\partial}{\partial z^j} \otimes d\bar{z}^k$$

is a $T^{2,0} \otimes T^{*0,1}$ tensor fields. The transformation rule is

$$T_{\bar{k}}^{ij} = \tilde{T}_{\bar{t}}^{rs} \frac{\partial z^i}{\partial \tilde{z}^r} \cdot \frac{\partial z^j}{\partial \tilde{z}^s} \cdot \frac{\partial \bar{\tilde{z}}^t}{\partial \bar{z}^k}.$$

6. Metrics

Define a **Hermitian metric** g on M to be a Hermitian inner product on the n-dimensional complex vector space $T_p^{1,0}M$ for each p, which varies smoothly in p.
Locally g is given by an $n \times n$ positive definite Hermitian matrix whose (i,j)-th entry we denote by $g_{i\bar{j}}$, which transforms according to

$$g_{i\bar{j}} = \tilde{g}_{k\bar{l}} \frac{\partial \tilde{z}^k}{\partial z^i} \frac{\partial \bar{\tilde{z}}^l}{\partial \bar{z}^j}, \quad \text{on } U \cap \tilde{U}.$$

Precisely, a Hermitian metric is a $T^{*1,1}$ tensor field satisfies:
- (Hermitian property) $\overline{g_{i\bar{j}}} = g_{j\bar{i}}$,
- (Positive definite) $g_{i\bar{j}} = X^i \bar{X}^j > 0$.

Given $T^{1,0}$ vector fields $X = X^i \frac{\partial}{\partial z^i}$ and $Y = Y^i \frac{\partial}{\partial z^i}$ we define their pointwise inner product by

$$\boxed{\langle X, Y \rangle_g = g_{i\bar{j}} X^i \bar{Y}^j,}$$

and we write

$$|X|_g = \sqrt{\langle X, X \rangle_g}$$

for the **norm** of X with respect to g.
Given $T^{*1,0}$ covector fields $a = a_i dz^i$ and $b = b^i dz^i$ we define their pointwise inner product by

$$\boxed{\langle a, b \rangle_g = g^{i\bar{j}} a_i \bar{b}_j,}$$

and we write

$$|a|_g = \sqrt{\langle a, a \rangle_g}$$

for the norm of a with respect to g.

Remark 6.1. *A Hermitian metric g defines a Riemannian metric g_R, which we can define locally by*

$$g_R\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right) = 2\mathrm{Re}(g_{i\bar{j}}) = g_R\left(\frac{\partial}{\partial y^i}, \frac{\partial}{\partial y^j}\right), \quad g_R\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial y^j}\right) = 2\mathrm{Im}(g_{i\bar{j}}).$$

Consider more general tensor. The set of locally defined functions $S^{\bar{i}}_{\bar{k}l} : U \longrightarrow \mathbb{C}$ defines a tensor , if it satisfies the transoformation rule:

$$S^{\bar{i}}_{k\bar{l}} = \tilde{S}^{\bar{a}}_{b\bar{c}} \frac{\partial \bar{z}^i}{\partial \bar{\tilde{z}}^a} \frac{\partial \tilde{z}^b}{\partial z^k} \frac{\partial \bar{\tilde{z}}^c}{\partial \bar{z}^l}, \quad \text{on } U \cap \tilde{U}.$$

More formally, $S = S^{\bar{i}}_{k\bar{l}} \frac{\partial}{\partial z^i} \otimes dz^k \otimes d\bar{z}^l$ defines a smooth section on $T^{0,1}M \otimes T^{*1,0}M \otimes T^{*0,1}M$. The norm of S is

$$|S|^2_g = S^{\bar{i}}_{k\bar{l}} S^{\bar{j}}_{r\bar{s}} g_{i\bar{j}} g^{k\bar{r}} g^{s\bar{l}}.$$

A Hermitian metric $g = (g_{i\bar{j}})$ is **Kähler** if

$$\boxed{\frac{\partial g_{i\bar{j}}}{\partial z^k} = \frac{\partial g_{k\bar{j}}}{\partial z^i}, \quad \text{for all } i, j, k = 1, \cdots, n.}$$

Example 3. $\mathbb{C}P^n$.
We can define metric

$$g_{i\bar{j}} = \partial_i \partial_{\bar{j}} \log(1^2 + |z_1|^2 + \cdots + |z_n|^2), \quad \text{on } U_0,$$

*and similarly for $U_1, \cdots, U_n$. This metric is called the **Fubini-Study metric**.*
We will show that $g_{i\bar{j}}$ defines a Kähler metric on $\mathbb{C}P^n$.
Hermitian property.
Since $\overline{\partial_i \partial_{\bar{j}}} = \partial_j \partial_{\bar{i}}$, then

$$\overline{g_{i\bar{j}}} = g_{j\bar{i}}.$$

Positive definite.

$$\begin{aligned} g_{i\bar{j}} &= \partial_i \left(\frac{\partial_{\bar{j}}(1 + |z_1|^2 + \cdots + |z_n|^2)}{1 + |z_1|^2 + \cdots + |z_n|^2} \right) \\ &= \partial_i \left(\frac{z_j}{1 + |z_1|^2 + \cdots + |z_n|^2} \right) \\ &= \frac{\delta_{ij}}{1 + |z_1|^2 + \cdots + |z_n|^2} - \frac{\bar{z}_i z_j}{(1 + |z_1|^2 + \cdots + |z_n|^2)^2}. \end{aligned}$$

Take $V \neq 0$, we have

$$\sum_{i,j=1}^n g_{i\bar{j}} V^i \bar{V}^j = \frac{\sum_{i=1}^n |V^i|^2}{1 + |z_1|^2 + \cdots + |z_n|^2} - \sum_{i,j=1}^n \frac{\bar{z}_i z_j V^i \bar{V}^j}{(1 + |z_1|^2 + \cdots + |z_n|^2)^2}.$$

For the second term, by Cauchy-Schwarz inequality, we obtain

$$\left(\sum_{i=1}^{n} \bar{z}_i V^i\right)\left(\sum_{j=1}^{n} z_j \bar{V}^j\right) \leq \sum_{i=1}^{n} |z_i|^2 \sum_{i=1}^{n} |V^i|^2 < (1+|z_1|^2+\cdots+|z_n|^2)\sum_{i=1}^{n}|V^i|^2.$$

Thus, $\sum_{i,j=1}^{n} g_{i\bar{j}} V^i \bar{V}^j \geq 0.$

<u>*Kähler.*</u>

$$\partial_k \partial_i \partial_{\bar{j}} \log(1+|z_1|^2+\cdots+|z_n|^2) = \partial_i \partial_k \partial_{\bar{j}} \log(1+|z_1|^2+\cdots+|z_n|^2).$$

Example 4. *Riemann surfaces.*
Any Hermitian metric is Kähler, by

$$\frac{\partial}{\partial z^1} g_{1\bar{1}} = \frac{\partial}{\partial z^1} g_{1\bar{1}}.$$

7. Covariant Differentiation

If we are given a $T^{1,0}$ vector field $X = X^i \partial_i$, the object

$$\frac{\partial X^i}{\partial \bar{z}^l},$$

givens a well-defined tensor, meaning that it transforms according to the rule

$$\frac{\partial X^i}{\partial \bar{z}^l} = \frac{\partial \tilde{X}^k}{\partial \bar{\tilde{z}}^j} \frac{\partial \bar{\tilde{z}}^j}{\partial \bar{z}^l} \frac{\partial z^i}{\partial \tilde{z}^k}, \quad \text{on } U \cap \tilde{U}.$$

Indeed, this follows from the fact that the transition maps for the X^i are holomorphic. However, the object

$$\frac{\partial X^i}{\partial z^k}$$

does not give a well-defined tensor.
We define the **Christoffel symbols** of g on the chart (U, z) to be the functions $\Gamma_{ij}^k : U \longrightarrow \mathbb{C}$ defined by

$$\boxed{\Gamma_{ij}^k = g^{k\bar{q}} \partial_i g_{j\bar{q}},}$$

where $g^{k\bar{q}}$ are the components of the inverse of g.
By the Kähler condition, we have

$$\Gamma_{ij}^k = \Gamma_{ji}^k.$$

Given a $T^{1,0}$ vector field X, we define the **covariant derivative** $\nabla_i X^j$ by

$$\boxed{\nabla_i X^j = \partial_i X^j + \Gamma_{ik}^j X^k,}$$

and this does define a tensor. By the above observation, $\partial_{\bar{l}} X^i$ is already a tensor and we define

$$\boxed{\nabla_{\bar{l}} X^i = \partial_{\bar{l}} X^i.}$$

Similarly, for a $T^{0,1}$ vector field $Y = Y^{\bar{j}} \partial_{\bar{j}}$, a $(1,0)$ form $a = a_i dz^i$ and a $(0,1)$ form $b = b_{\bar{j}} d\bar{z}^j$, we define

$$\boxed{\begin{aligned}
\nabla_k Y^{\bar{j}} &= \partial_k Y^{\bar{j}} \\
\nabla_{\bar{l}} Y^{\bar{j}} &= \partial_{\bar{l}} Y^{\bar{j}} + \overline{\Gamma^j_{lq}} Y^{\bar{q}} \\
\nabla_k a_i &= \partial_k a_i - \Gamma^p_{ki} a_p \\
\nabla_{\bar{l}} a_i &= \partial_{\bar{l}} a_i \\
\nabla_k b_{\bar{j}} &= \partial_k b_{\bar{j}} \\
\nabla_{\bar{l}} b_{\bar{j}} &= \partial_{\bar{l}} b_{\bar{j}} - \overline{\Gamma^q_{lj}} b_{\bar{q}}.
\end{aligned}}$$

Moreover, we can extend covariant differentiation naturally to any kind of tensor, such as the $T^j_{\bar{k}lp}$, we have

$$\begin{aligned}
\nabla_i T^j_{\bar{k}lp} &= \partial_i T^j_{\bar{k}lp} + \Gamma^j_{iq} T^q_{\bar{k}lp} - \Gamma^q_{il} T^i_{\bar{k}qp} - \Gamma^q_{ip} T^j_{\bar{k}lq} \\
\nabla_{\bar{i}} T^j_{\bar{k}lp} &= \partial_{\bar{i}} T^j_{\bar{k}lp} - \Gamma^q_{ik} T^i_{\bar{q}lp}.
\end{aligned}$$

In particular, we have

$$\boxed{\nabla_i g_{j\bar{k}} = 0.}$$

Indeed,

$$\begin{aligned}
\nabla_i g_{j\bar{k}} &= \partial_i g_{j\bar{k}} - \Gamma^l_{ij} g_{l\bar{k}} \\
&= \partial_i g_{j\bar{k}} - g^{l\bar{q}} (\partial_i g_{j\bar{q}}) g_{l\bar{k}} \\
&= \partial_i g_{j\bar{k}} - \partial_i g_{j\bar{k}} = 0.
\end{aligned}$$

8. Curvature

scribe the curvature associated to a Kähler metric g.
Define the **curvature tensor** $R^j_{ik\bar{l}}$ by

$$\boxed{R^j_{ik\bar{l}} = -\partial_{\bar{l}} \Gamma^j_{ik}.}$$

It will be convenient to define

$$\boxed{R_{i\bar{j}k\bar{l}} = g_{p\bar{j}} R^p_{ik\bar{l}}.}$$

• **Ricci curvature**

$$\boxed{R_{i\bar{j}} = g^{k\bar{l}} R_{i\bar{j}k\bar{l}}}$$

• **Scalar curvature**

$$\boxed{R = g^{i\bar{j}} R_{i\bar{j}}.}$$

We now show the explicit formula for curvature

$$\begin{aligned} R_{i\bar{j}k\bar{l}} &= g_{p\bar{j}} R^p_{ik\bar{l}} = -g_{p\bar{j}}(\partial_{\bar{l}} \Gamma^p_{ik}) = -g_{p\bar{j}} \partial_{\bar{l}}(g^{p\bar{q}} \partial_i g_{k\bar{q}}) \\ &= -g_{p\bar{j}} g^{p\bar{q}} \partial_{\bar{l}} \partial_i g_{k\bar{q}} + g_{p\bar{j}}(g^{p\bar{s}} g^{r\bar{q}} \partial_{\bar{l}} g_{r\bar{s}}) \partial_i g_{k\bar{q}} \\ &= -\partial_i \partial_{\bar{l}} g_{k\bar{j}} + g^{r\bar{q}}(\partial_{\bar{l}} g_{r\bar{j}}) \partial_i g_{k\bar{q}}. \end{aligned}$$

We obtain the formula

$$\boxed{R_{i\bar{j}k\bar{l}} = -\partial_i \partial_{\bar{l}} g_{k\bar{j}} + g^{r\bar{q}}(\partial_{\bar{l}} g_{r\bar{j}}) \partial_i g_{k\bar{q}}.}$$

Note that the fourth identity by

$$dA^{-1} = -A^{-1}(dA)A^{-1},$$

where the A^{-1} is the inverse matrix of A.
Since g is Kähler metric, by the Kähler condition, we have

$$\boxed{R_{i\bar{j}k\bar{l}} = R_{k\bar{j}i\bar{l}} = R_{i\bar{l}k\bar{j}} = R_{k\bar{l}i\bar{j}}.}$$

The curvature tensor measures the failure of covariant derivatives to commute.
More precisely:
For a $T^{1,0}$ vector field $X = X^p \partial_p$, a $T^{0,1}$ vector field $Y = Y^{\bar{q}} \partial_{\bar{q}}$, a $(1,0)$ form $a = a_p dz^p$ and a $(0,1)$ form $b = b_{\bar{q}} d\bar{z}^q$, we have the following commutation formulas:

$$\boxed{\begin{aligned} [\nabla_i, \nabla_{\bar{j}}] X^p &= R^p_{li\bar{j}} X^l = R_{l\bar{k}i\bar{j}} g^{p\bar{k}} X^l \\ [\nabla_i, \nabla_{\bar{j}}] Y^{\bar{q}} &= -R^{\bar{q}}_{\bar{l}i\bar{j}} Y^{\bar{l}} = -R_{k\bar{l}i\bar{j}} g^{k\bar{q}} Y^{\bar{l}} \\ [\nabla_i, \nabla_{\bar{j}}] a_p &= -R^k_{pi\bar{j}} a_k = -R_{p\bar{l}i\bar{j}} g^{k\bar{l}} a_k \\ [\nabla_i, \nabla_{\bar{j}}] b_{\bar{q}} &= R^{\bar{k}}_{\bar{q}i\bar{j}} b_{\bar{k}} = R_{l\bar{q}i\bar{j}} g^{l\bar{k}} b_{\bar{k}}, \end{aligned}}$$

where $[\nabla_i, \nabla_{\bar{j}}] = \nabla_i \nabla_{\bar{j}} - \nabla_{\bar{j}} \nabla_i$.
We will show the first formlua.

Proof.

$$\begin{aligned}
[\nabla_i, \nabla_{\bar{j}}]X^p &= \nabla_i(\nabla_{\bar{j}}X^p) - \nabla_{\bar{j}}(\nabla_i X^p) \\
&= \nabla_i(\partial_{\bar{j}}X^p) - \nabla_{\bar{j}}(\partial_i X^p + \Gamma^p_{il}X^l) \\
&= \partial_i\partial_{\bar{j}}X^p + \Gamma^p_{il}\partial_{\bar{j}}X^l - \partial_{\bar{j}}\partial_i X^p - \partial_{\bar{j}}\Gamma^p_{il}X^l - \Gamma^p_{il}\partial_{\bar{j}}X^l \\
&= -(\partial_{\bar{j}}\Gamma^p_{il})X^l = -R^p_{li\bar{j}}X^l.
\end{aligned}$$

□

We now give a formula for the Ricci curvature.
Precisely, Ricci curvature can be written by

$$\boxed{R_{i\bar{j}} = -\partial_i\partial_{\bar{j}} \log\det g.}$$

Proof. Recall: Assume $A_{i\bar{j}} := A_{i\bar{j}}(t)$ is an invertible Hermitian matrix, then

$$\frac{d}{dt}\det A = \frac{d}{dt}\left(\sum_j A_{i\bar{j}}M_{i\bar{j}}\right),$$

where $M_{i\bar{j}}$ is the cofactor of the (i,j) entry. Notice that $M_{i\bar{j}}$ does not depend on the (i,j) entry, thus

$$\frac{d}{dt}\det A = \frac{\partial}{\partial A_{i\bar{j}}}\det A\frac{d}{dt}A_{i\bar{j}} = M_{i\bar{j}}\frac{d}{dt}A_{i\bar{j}}.$$

Recall: $A^{i\bar{j}} = \dfrac{M_{i\bar{j}}}{\det A}$, then

$$\frac{d}{dt}\det A = \det A \cdot A^{i\bar{j}} \cdot \frac{d}{dt}A_{i\bar{j}}.$$

Therefore,

$$\frac{d}{dt}\log\det A = A^{i\bar{j}}\frac{d}{dt}A_{i\bar{j}}.$$

Thus,

$$\begin{aligned}
-\partial_i\partial_{\bar{j}}\log\det g &= -\partial_i(g^{k\bar{l}}\partial_{\bar{j}}g_{k\bar{l}}) \\
&= -g^{k\bar{l}}\partial_i\partial_{\bar{j}}g_{k\bar{l}} + g^{k\bar{s}}g^{r\bar{l}}(\partial_i g_{r\bar{s}})(\partial_{\bar{j}}g_{k\bar{l}}) \\
&= g^{k\bar{l}}R_{i\bar{j}k\bar{l}} = R_{i\bar{j}}.
\end{aligned}$$

□

9. Differential Forms

A (p,q) form χ is a tensor of type $T^{*p,q}$ of the form

$$\chi = \frac{1}{p!q!}\chi_{i_1\cdots i_p\bar{j}_1\cdots\bar{j}_q}dz^{i_1}\wedge\cdots\wedge dz^{i_p}\wedge d\bar{z}^{j_1}\wedge\cdots d\bar{z}^{j_q},$$

where $\chi_{i_1\cdots i_p\bar{j}_1\cdots\bar{j}_q}$ is anti-symmetric in $i_1,\cdots,i_p$ and $\bar{j}_1,\cdots,\bar{j}_q$.
e.g.

$$\chi_{i_1i_2\cdots i_p} = -\chi_{i_2i_1\cdots i_p}.$$

We define two operators ∂ and $\bar{\partial}$.

$$\begin{aligned}\partial: &\quad (p,q)\text{ form}\longrightarrow(p+1,q)\text{ form},\\ \bar{\partial}: &\quad (p,q)\text{ form}\longrightarrow(p,q+1)\text{ form}.\end{aligned}$$

Precisely,

$$\partial\chi = \frac{1}{p!q!}\partial_k\chi_{i_1\cdots i_p\bar{j}_1\cdots\bar{j}_q}dz^k\wedge dz^{i_1}\wedge\cdots\wedge dz^{i_p}\wedge d\bar{z}^{j_1}\wedge\cdots\wedge d\bar{z}^{j_q},$$

and

$$\bar{\partial}\chi = \frac{1}{p!q!}\partial_{\bar{k}}\chi_{i_1\cdots i_p\bar{j}_1\cdots\bar{j}_q}d\bar{z}^k\wedge dz^{i_1}\wedge\cdots\wedge dz^{i_p}\wedge d\bar{z}^{j_1}\wedge\cdots\wedge d\bar{z}^{j_q}.$$

We write

$$d = \partial + \bar{\partial},$$

is called **exterior derivative**. We have the following facts that

$$\partial^2 = \bar{\partial}^2 = d^2 = 0, \quad \partial\bar{\partial} = -\bar{\partial}\partial.$$

Define the **Cohomology groups** $H^{p,q}_{\bar{\partial}}(M)$ by

$$\begin{aligned}H^{p,q}_{\bar{\partial}}(M) &= \frac{\{\varphi\in(p,q)-\text{forms}\mid\bar{\partial}\varphi=0\}}{\{\chi\in(p,q)-\text{forms}\mid\chi=\bar{\partial}\eta,\text{some }(p,q-1)\text{-forms }\eta\}}\\ &= \frac{\{\bar{\partial}-\text{closed }(p,q)\text{-forms}\}}{\{\bar{\partial}-\text{exact }(p,q)\text{-forms}\}}.\end{aligned}$$

Given a Kähler metric g, then the associated **Kähler form** ω is

$$\boxed{\omega = \sqrt{-1}g_{i\bar{j}}dz^i\wedge d\bar{z}^j.}$$

Since $g_{i\bar{j}}$ is Hermitian, the form ω is real. Indeed,

$$\bar{\omega} = -\sqrt{-1}\overline{g_{i\bar{j}}}d\bar{z}^i\wedge dz^j = \sqrt{-1}g_{j\bar{i}}dz^j\wedge d\bar{z}^i = \omega.$$

We have the following property that

$$g\text{ in Kähler}\Longleftrightarrow d\omega=0\Longleftrightarrow\partial\omega=0\Longleftrightarrow\bar{\partial}\omega=0$$

Proof. It is obvious that

$$d\omega = 0 \Longrightarrow \partial\omega = 0 \Longrightarrow \bar{\partial}\omega = 0$$

We will check $\partial_k g_{i\bar{j}} = \partial_i g_{k\bar{j}} \Longrightarrow \partial\omega = 0$.
By direct calculation, we obtain

$$\begin{aligned}
\partial\omega &= \sqrt{-1} g_{i\bar{j}k} dz^k \wedge dz^i \wedge d\bar{z}^j \\
&= \sqrt{-1} \sum_{i \neq k} g_{i\bar{j}k} dz^k \wedge dz^i \wedge d\bar{z}^j \\
&= \sqrt{-1} \sum_{i<k} g_{i\bar{j}k} dz^k \wedge dz^i \wedge d\bar{z}^j + \sqrt{-1} \sum_{i>k} g_{i\bar{j}k} dz^k \wedge dz^i \wedge d\bar{z}^j \\
&= \sqrt{-1} \sum_{i<k} g_{i\bar{j}k} dz^k \wedge dz^i \wedge d\bar{z}^j + \sqrt{-1} \sum_{i<k} g_{k\bar{j}i} dz^i \wedge dz^k \wedge d\bar{z}^j \\
&= \sqrt{-1} \sum_{i<k} g_{i\bar{j}k} dz^k \wedge dz^i \wedge d\bar{z}^j - \sqrt{-1} \sum_{i>k} g_{k\bar{j}i} dz^k \wedge dz^i \wedge d\bar{z}^j,
\end{aligned}$$

then

$$\partial_k g_{i\bar{j}} = \partial_i g_{k\bar{j}} \Longrightarrow \partial\omega = 0.$$

□

Briefly, $\sqrt{-1} g_{i\bar{j}k} dz^k \wedge dz^i \wedge d\bar{z}^j$, $g_{i\bar{j}k}$ symmetric in i, k and $dz^k \wedge dz^i \wedge d\bar{z}^j$ anti-symmetric in i, k, then $\partial\omega = 0$.
If ω is a Kähler form $[\omega] \in H^{p,q}_{\bar{\partial}}(M)$, i.e., cohomology class of ω. We define the **Ricci form** of g to be the $(1,1)$ form

$$\boxed{\mathrm{Ric}(\omega) = \sqrt{-1} R_{i\bar{j}} dz^i \wedge d\bar{z}^j = -\sqrt{-1}\partial\bar{\partial} \log\det g.}$$

$\mathrm{Ric}(\omega)$ is a real closed $(1,1)$-form.

Proof. Since

$$\begin{aligned}
\overline{\mathrm{Ric}(\omega)} &= -\sqrt{-1}(\overline{-\partial_i \partial_{\bar{j}} \log\det g}) d\bar{z}^i \wedge dz^j \\
&= -\sqrt{-1}\partial_j \partial_{\bar{i}} \log\det g dz^j \wedge d\bar{z}^i = \mathrm{Ric}(\omega),
\end{aligned}$$

then Ricci form is real.

$$\partial\mathrm{Ric}(\omega) = -\sqrt{-1}\partial_k \partial_i \partial_{\bar{j}} \log\det g dz^k \wedge dz^i \wedge d\bar{z}^j = 0,$$

since $\partial_k \partial_i \partial_{\bar{j}}$ is symmetric in i, k and $dz^k \wedge dz^i \wedge d\bar{z}^j$ is anti-symmetric in i, k.
Similarly, $\bar{\partial}\mathrm{Ric}(\omega) = 0$ and $d\mathrm{Ric}(\omega) = 0$. □

We define the **first Chern class** by

$$c_1(M) = \frac{1}{\pi}[\mathrm{Ric}(\omega)] \in H^{1,1}_{\bar{\partial}}(M, \mathbb{R}) = \frac{\bar{\partial}\text{-closed real } (1,1) \text{ forms}}{\bar{\partial}\text{-exact real } (1,1) \text{ forms}}.$$

Note that $c_1(M) \in H^{1,1}_{\bar\partial}(M, \mathbb{R})$ is independent of choice of ω.

10. Holomorphic Line Bundles

Rough idea: A line bundle L consists of a copy of $\mathbb{C}$ over every point in M, locally,

$$L \cong U_\alpha \times \mathbb{C}.$$

Precisely,
A **holomorphic line bundle** L over M is

$$L = \bigsqcup_{x \in M} L_x,$$

where L_x is a complex vector space of dimension 1. Endowed with the structure of a complex manifold and there is a holomorphic map(projection map)

$$\begin{aligned} \pi : \quad L &\longrightarrow M \\ L_x &\longmapsto \{x\}, \end{aligned}$$

such that for every point $x \in M$ there is an open set $x \in U_\alpha$ and a biholomorphic map

$$\psi_\alpha : \pi^{-1}(U_\alpha) \longrightarrow U_\alpha \times \mathbb{C}$$

with ψ_α mapping L_x to $\mathbb{C}$ isomorphically.
Given a line bundle L and $U_\alpha \cap U_\beta \neq \emptyset$ define transition functions

$$t_{\alpha\beta} : \quad U_\alpha \cap U_\beta \longrightarrow \mathbb{C} \setminus \{0\}$$

holomorphc by

$$t_{\alpha\beta}(x) = (\psi_\alpha \circ \psi_\beta^{-1}) \mid_{\{x\} \times \mathbb{C}} .$$

They satisfy
(1) $t_{\alpha\beta} \circ t_{\beta\alpha} = 1$ on $U_\alpha \cap U_\beta$,
(2)$t_{\alpha\beta} \circ t_{\beta\gamma} \circ t_{\gamma\alpha} = 1$ on $U_\alpha \cap U_\beta \cap U_\gamma$.
Given tansition functions $t_{\alpha\beta} : U_\alpha \cap U_\beta \longrightarrow \mathbb{C} \setminus \{0\}$ holomorphic satisfying (1)&(2). We can define a **holomorphic line bundle**

$$L = \bigsqcup U_\alpha \times \mathbb{C} / \sim,$$

where $(x, \lambda) \in U_\beta \times \mathbb{C} \sim (x, t_{\alpha\beta}\lambda) \in U_\alpha \times \mathbb{C}$.
Thus, we will think about a line bundle as a set of transition functions $\{t_{\alpha\beta}\}$.

Example 5. *$M \times \mathbb{C}$ is trivial line bundle.*
$\{U_\alpha\}$ is any open cover and $t_{\alpha\beta} = 1$.

Example 6. *$\mathbb{C}P^n$.*
Take $U_\alpha = \{[z_0, z_1, \cdots, z_n] \mid z_\alpha \neq 0\}$ and define $t_{\alpha\beta} = \frac{z_\beta}{z_\alpha} \neq 0$ on $U_\alpha \cap U_\beta$ holomorphic.
• $t_{\alpha\beta} \circ t_{\beta\alpha} = \frac{z_\beta}{z_\alpha} \cdot \frac{z_\alpha}{z_\beta} = 1$,

• $t_{\alpha\beta} \circ t_{\beta\gamma} \circ t_{\gamma\alpha} = \frac{z_\beta}{z_\alpha} \cdot \frac{z_\gamma}{z_\beta} \cdot \frac{z_\alpha}{z_\gamma} = 1.$
It is called the ***hyperplane line bundle*** *H.*

A holomorphic **section** of a line bundle L over M is a map

$$s : M \longrightarrow L$$

holomorphic satisfying

$$\pi \circ s = \mathrm{id}_M.$$

In terms of transition functions, define local holomorphic function

$$\begin{aligned} s_\alpha : &\quad U_\alpha \longrightarrow \mathbb{C} \\ s_\beta : &\quad U_\beta \longrightarrow \mathbb{C} \end{aligned}$$

by

$$\psi_\alpha(s(x)) = (x, s_\alpha(x)) \in U_\alpha \times \mathbb{C}$$

and

$$\psi_\beta(s(x)) = (x, s_\beta(x)) \in U_\beta \times \mathbb{C}.$$

On $U_\alpha \cap U_\beta$ how does $s_\alpha(x)$ relate to $s_\beta(x)$?
Since $\psi_\alpha(s(x)) = (x, s_\alpha(x))$ and $\psi_\beta(s(x)) = (x, s_\beta(x))$, then $s(x) = \psi_\beta^{-1}(x, s_\beta(x))$ and

$$(x, s_\alpha(x)) = \psi_\alpha \circ \psi_\beta^{-1}(x, s_\beta(x)).$$

Notice that $\psi_\alpha \circ \psi_\beta^{-1} = t_{\alpha\beta}$ restrict to $\{x\} \times \mathbb{C}$.
Thus the relation is

$$s_\alpha(x) = t_{\alpha\beta}(x) s_\beta(x).$$

Given a line bundle $\{t_{\alpha\beta}\}$, think of a section s as a collection of holomorphic functions $\{s_\alpha\}$, where $s_\alpha : U_\alpha \longrightarrow \mathbb{C}$ satisfying

$$s_\alpha = t_{\alpha\beta} s_\beta, \quad \text{on } U_\alpha \cap U_\beta.$$

Example 7. *H: Hyperplane line bundle. Transition function is $t_{\alpha\beta} = \frac{z_\beta}{z_\alpha}$.*
Consider $s_\alpha = \frac{z_0}{z_\alpha}$ on U_α. This is a holomorphic section of H. Indeed,

$$s_\alpha = \frac{z_0}{z_\alpha} = \frac{z_\beta}{z_\alpha} \cdot \frac{z_0}{z_\beta} = t_{\alpha\beta} s_\beta, \quad \text{on } U_\alpha \cap U_\beta.$$

Example 8. *Consider H^l ($l = 1, 2, 3, \cdots$) means that the line bundle with $t_{\alpha\beta} = \left(\frac{z_\beta}{z_\alpha}\right)^l$.*
Take P to be a homogeneous polynomial of degree l in $n+1$ variables

$$P(\lambda x_0, \lambda x_1, \cdots, \lambda x_n) = \lambda^l P(x_0, x_1, \cdots, x_n).$$

Consider $s_\alpha = p\left(\frac{z_0}{z_\alpha}, \frac{z_1}{z_\alpha}, \cdots, \frac{z_n}{z_\alpha}\right)$. *This is a holomorphic section of* H^l. *Indeed,*

$$\begin{aligned} s_\alpha &= P\left(\frac{z_0}{z_\alpha}, \frac{z_1}{z_\alpha}, \cdots, \frac{z_n}{z_\alpha}\right) \\ &= P\left(\frac{z_0}{z_\beta}\cdot\frac{z_\beta}{z_\alpha}, \frac{z_1}{z_\beta}\cdot\frac{z_\beta}{z_\alpha}, \cdots, \frac{z_n}{z_\beta}\cdot\frac{z_\beta}{z_\alpha}\right) \\ &= \left(\frac{z_\beta}{z_\alpha}\right)^l P\left(\frac{z_0}{z_\beta}, \frac{z_1}{z_\beta}, \cdots, \frac{z_n}{z_\beta}\right) \\ &= t_{\alpha\beta} s_\beta. \end{aligned}$$

We consider Hermitian metrics on line bundles.
Given a section s, we would like to know how to measure it at a point,

$$|s|^2 = h_\alpha s_\alpha \overline{s_\alpha},$$

where $\{h_\alpha\}$ is a set of positive smooth function $h_\alpha : U_\alpha \longrightarrow (0, \infty)$. Call this collection h. Pick the h_α so that $h_\alpha s_\alpha \overline{s_\alpha}$ is independent of patch U_α ,i.e.,

$$h_\alpha s_\alpha \overline{s_\alpha} = h_\beta s_\beta \overline{s_\beta}, \quad \text{on } U_\alpha \cap U_\beta.$$

Since

$$h_\alpha s_\alpha \overline{s_\alpha} = h_\alpha t_{\alpha\beta} s_\beta \overline{t_{\alpha\beta} s_\beta} = h_\alpha |t_{\alpha\beta}|^2 s_\beta \overline{s_\beta},$$

then we need $h_\alpha |t_{\alpha\beta}|^2 = h_\beta$ transformation rule for a Hermitian metric.

Example 9. *H: Hyperplane line bundle. Transition function is* $t_{\alpha\beta} = \frac{z_\beta}{z_\alpha}$.
A metric might be

$$h_\alpha = \left(\left|\frac{z_0}{z_\alpha}\right|^2 + \cdots + \left|\frac{z_n}{z_\alpha}\right|^2\right)^{-1}.$$

Indeed,

$$h_\alpha |t_{\alpha\beta}|^2 = \left(\left|\frac{z_0}{z_\beta}\right|^2 + \cdots + \left|\frac{z_n}{z_\beta}\right|^2\right)^{-1} = h_\beta.$$

Example 10. *Let M be any complex manifold. Let $\{U_\alpha\}$ be a covering of coordinate charts with z^i. Define*

$$t_{\alpha\beta} = \det\left(\frac{\partial \tilde{z}^i}{\partial z^j}\right).$$

Note that $t_{\alpha\beta} : U_\alpha \cap U_\beta \longrightarrow \mathbb{C} \setminus \{0\}$ holomorphic. This gives the ***canonical line bundle*** *K.*
Also have K^{-1}: anti-canonical line bundle.
Given any line bundle: L with transition functions $\{t_{\alpha\beta}\}$. The inverse line bundle L^{-1} is the line bundle with transition functions $\{t_{\alpha\beta}^{-1}\}$.
The holomorphic sections of K can be thought of as holomorphic $(n, 0)$-forms.

On U_α, let $\mu = \frac{1}{n!}\mu_{i_1\cdots i_n} dz^{i_1}\wedge\cdots\wedge dz^{i_n}$ with $\partial_k \mu_{i_1\cdots i_n} = 0$ be a holomorphic $(n,0)$-forms. By the following property

$$d\tilde{z}^1\wedge\cdots\wedge d\tilde{z}^n = \det\left(\frac{\partial \tilde{z}^j}{\partial z^i}\right) dz^1\wedge\cdots\wedge dz^n,$$

we have

$$\begin{aligned}\mu &= \mu_{1\cdots n} dz^1\wedge\cdots\wedge dz^n\\ &= \tilde{\mu}_{1\cdots n} d\tilde{z}^1\wedge\cdots\wedge d\tilde{z}^n\\ &= \tilde{\mu}_{1\cdots n}\det\left(\frac{\partial \tilde{z}^j}{\partial z^i}\right) dz^1\wedge\cdots\wedge dz^n.\end{aligned}$$

Thus, $\mu_{1\cdots n}$ transform as $\mu_{1\cdots n} = \det\left(\frac{\partial \tilde{z}^j}{\partial z^i}\right)\tilde{\mu}_{1\cdots n}$, i.e.,

$$s_\alpha = t_{\alpha\beta} s_\beta.$$

11. Coordinate Systems

Suppose (M,g) is Kähler manifold. For any point $p\in M$, there exist local coordinate systems $z^1,\cdots,z^n$ centered at p, i.e., $z^i(p)=0$, $\forall i$, such that

$$g_{i\bar{j}}(0) = \delta_{ij}, \quad \partial_k g_{i\bar{j}}(0) = 0.$$

Proof. See [[2] Proposition 1.16]. □

12. Integration

Suppose (M,g) is compact Kähler manifold, ω is Kähler form of g. On a Kähler manifold, there is a natural volume form

$$\boxed{\frac{\omega^n}{n!} = \left(\frac{\sqrt{-1}}{2}\right)^n \det g\, dz^1\wedge d\bar{z}^1\wedge\cdots\wedge dz^n\wedge d\bar{z}^n.}$$

Introduce a key tool is **Stoke's Theorem**, i.e.,

$$\boxed{\int_M d\eta = 0, \quad \text{if } \eta \text{ is a } (2n-1) \text{ form.}}$$

In particular,

$$\boxed{\int_M \partial\eta = 0, \quad \text{if } \eta \text{ is a } (n-1,n) \text{ form}}$$

and

$$\boxed{\int_M \bar{\partial}\eta = 0, \quad \text{if } \eta \text{ is a } (n,n-1) \text{ form.}}$$

We consider the **Laplacian** on functions. If g is a Kähler metric, write

$$\boxed{\Delta_g f = g^{i\bar{j}}\partial_i\partial_{\bar{j}} f}$$

for f is a function on M.
We have a corollary of Stoke's theorem that

$$\boxed{\int_M \Delta_g f \omega_n = 0.}$$

Proof. Recall a basic formula: For any $(1,1)$ form $\eta = \frac{\sqrt{-1}}{2}\eta_{i\bar{j}} dz^i \wedge d\bar{z}^j$, we have

$$\boxed{(g^{i\bar{j}}\eta_{i\bar{j}})\omega^n = n\eta \wedge \omega^{n-1}}$$

Moreover

$$\boxed{(g^{i\bar{j}}\partial_i\partial_{\bar{j}} f)\omega^n = \frac{\sqrt{-1}}{2} n\partial\bar{\partial} f \wedge \omega^{n-1}.}$$

proof of the basic formula.
Pick coordinate system at a point so that

$$\omega = \frac{\sqrt{-1}}{2}\sum_{i=1}^n dz^i \wedge d\bar{z}^i.$$

$$\begin{aligned}
\text{RHS} &= n(n-1)!\left(\frac{\sqrt{-1}}{2}\right)^{n-1}\sum_{k=1}^n\sum_{i\neq j}^n dz^1\wedge d\bar{z}^1 \wedge \cdots \wedge \widehat{dz^k \wedge d\bar{z}^k} \wedge \cdots \wedge dz^n \wedge d\bar{z}^n \wedge \frac{\sqrt{-1}}{2}\eta_{i\bar{j}} dz^i \wedge d\bar{z}^j \\
&= n!\left(\frac{\sqrt{-1}}{2}\right)^n (\eta_{1\bar{1}} + \cdots + \eta_{n\bar{n}}) dz^1 \wedge d\bar{z}^1 \wedge \cdots \wedge dz^n \wedge d\bar{z}^n \\
&= (g^{i\bar{j}}\eta_{i\bar{j}})\omega^n = \text{LHS}
\end{aligned}$$

we obtain the basic formula.
Then, by the basic formula, we have

$$\int_M \Delta_g f \omega^n = n\int_M \frac{\sqrt{-1}}{2}\partial\bar{\partial} f \wedge \omega^{n-1} = n\int_M \partial\left(\frac{\sqrt{-1}}{2}\bar{\partial} f \wedge \omega^{n-1}\right) = 0.$$

The third equality by the fact that $\partial\omega = 0$ and the last equality by the Stoke's theorem. $\square$

We have the following regularity result about Laplace equation (Basic Hodge theory).
If $v \in C^\infty(M)$ with $\int_M v\omega^n = 0$, then exists $f \in C^\infty(M)$ with $\Delta_g f = v$ and f is unique up to addition of a constant.
We will prove the uniqueness.

Proof. Suppose f_1, f_2 are solutions of $\Delta_g f = v$, i.e., $\Delta_g f_1 = \Delta_g f_2$. Write $\psi = f_1 - f_2$, then

$$\int_M \psi \Delta_g \psi \omega^n = 0.$$

By the Stoke's theorem, we have

$$\begin{aligned} &\int_M \psi \frac{\sqrt{-1}}{2} \partial\bar{\partial}\psi \wedge \omega^{n-1} = 0 \\ \Longrightarrow &\int_M \frac{\sqrt{-1}}{2} \partial\psi \wedge \bar{\partial}\psi \wedge \omega^{n-1} = 0 \\ \Longrightarrow &\frac{1}{n}\int_M |\nabla\psi|^2 \omega^n = 0, \end{aligned}$$

where

$$|\nabla\psi|^2 = g^{i\bar{j}} \partial_i \psi \partial_{\bar{j}} \psi.$$

Thus, $|\nabla\psi| = 0$ and ψ is constant. □

13. Curvature of Hermitian Metrics, Chern Class

Recall: Line bundle L is a set of transition functions $\{t_{\alpha\beta}\}$. A **Hermitian metric** h is a set of smooth functions $\{h_\alpha\}$ satisfying

$$h_\beta = |t_{\alpha\beta}|^2 h_\alpha, \quad \text{on } U_\alpha \cap U_\beta.$$

Define **curvature** ρ_h of h to be

$$\rho_h = -\frac{\sqrt{-1}}{2} \partial\bar{\partial} \log h_\alpha.$$

It is well-defined, i.e., we will check

$$-\frac{\sqrt{-1}}{2} \partial\bar{\partial} \log h_\alpha = -\frac{\sqrt{-1}}{2} \partial\bar{\partial} \log h_\beta.$$

Proof. By condition, we have

$$-\frac{\sqrt{-1}}{2} \partial\bar{\partial} \log h_\beta = -\frac{\sqrt{-1}}{2} \partial\bar{\partial} \log(h_\alpha |t_{\alpha\beta}|^2) = -\frac{\sqrt{-1}}{2} \partial\bar{\partial} \log h_\alpha - \frac{\sqrt{-1}}{2} \partial\bar{\partial} \log |t_{\alpha\beta}|^2.$$

The last term is zero, since we have the claim that f is non zero holomorphic function, then $\partial\bar{\partial} \log |f|^2 = 0$.
Indeed,

$$\begin{aligned} \partial_i \partial_{\bar{j}} \log |f|^2 &= \partial_i \left(\frac{\partial_{\bar{j}} |f|^2}{|f|^2} \right) = \partial_i \left(\frac{f \partial_{\bar{j}} \bar{f}}{f \bar{f}} \right) \\ &= \partial_i \left(\frac{\partial_{\bar{j}} \bar{f}}{\bar{f}} \right) = \frac{\partial_i \partial_{\bar{j}} \bar{f}}{\bar{f}} = 0, \end{aligned}$$

where the second identity by the fact that $\partial_{\bar{j}} f = 0$ (f is holomorphic function). □

Consider curvature form

$$\rho_h = -\frac{\sqrt{-1}}{2}\partial\bar{\partial}\log h$$

is closed 2 form.
Define the **first Chern class** of L to be

$$c_1(L) = \frac{1}{\pi}[\rho_h] \in H^{1,1}_{\bar{\partial}}(M, \mathbb{R}).$$

Cohomology class $[\rho_h]$ is independent of choice of h. Indeed

Proof. We want to show that if h' is another metric on L, then

$$\rho_h = \rho_{h'} + \bar{\partial}\eta,$$

where η is some $(1,0)$-form. Note $\frac{h'}{h} = e^f$ for some $f \in C^\infty(M)$. Thus,

$$\rho_{h'} = -\frac{\sqrt{-1}}{2}\partial\bar{\partial}\log h' = -\frac{\sqrt{-1}}{2}\partial\bar{\partial}\log he^f = \rho_h - \frac{\sqrt{-1}}{2}\partial\bar{\partial}f = \rho_h + \bar{\partial}\left(\frac{\sqrt{-1}}{2}\partial f\right)$$

and

$$[\rho_{h'}] = [\rho_h].$$

□

Recall the canonical line bundle K, anti-canonical line bundle K^{-1} and the transition functions

$$t_{\alpha\beta} = \det\left(\frac{\partial \tilde{z}^i}{\partial z^j}\right).$$

We can show that $\det g$ is a Hermitian metric on K^{-1}.
Then

$$c_1(K^{-1}) = \frac{1}{\pi}\left[-\frac{\sqrt{-1}}{2}\partial\bar{\partial}\log\det g\right] = \frac{1}{\pi}[\mathrm{Ric}(\omega)] = c_1(M).$$

We say that cohomology class $\alpha \in H^{1,1}_{\bar{\partial}}(M, \mathbb{R})$ is **positive** (write $\alpha > 0$) if there is a Kähler form $\omega \in \alpha$ (i.e., $[\omega] = \alpha$).

We give a important lemma that **$\partial\bar{\partial}$-Lemma**.
Suppose M is a compact Kähler manifold. Let η be a $\bar{\partial}$-closed real $(1,1)$ form which is $\bar{\partial}$-exact, then there exist $f \in C^\infty(M)$ unique up to addition of a constant such that

$$\eta = \frac{\sqrt{-1}}{2}\partial\bar{\partial}f.$$

By $\partial\bar{\partial}$-Lemma, we have the following result.
At most one of the following can hold for $\alpha \in H^{1,1}_{\bar{\partial}}(M, \mathbb{R})$, (1) $\alpha > 0$, (2) $\alpha = 0$, (3) $\alpha < 0$. If none of those hold we say α is indefinite.

Proof. Suppose (1) and (2) hold, that is $\alpha > 0$, i.e., $\alpha = [\omega]$ and $\alpha = 0$, i.e., $\alpha = [0]$. Then

$$\omega = 0 + \bar{\partial}\psi$$

for some $(1,0)$ form ψ.
By $\partial\bar{\partial}$-Lemma, we have $\omega = \sqrt{-1}\partial\bar{\partial}f$ for some function f and $g_{i\bar{j}} = \partial_i\partial_{\bar{j}}f$. Then,

$$n = g^{i\bar{j}}g_{i\bar{j}} = g^{i\bar{j}}\partial_i\partial_{\bar{j}}f = \Delta_g f.$$

Therefore,

$$0 = \int_M \Delta_g f\omega^n = \int_M n\omega^n > 0.$$

□

If $\alpha > 0$ ($\alpha \in H^{1,1}_{\bar{\partial}}(M,\mathbb{R})$), we say α is a **Kähler class**, write $\alpha = [\omega]$.
By $\partial\bar{\partial}$-Lemma, any Kähler form in $[\omega]$ can be written as

$$\tilde{\omega} = \omega + \sqrt{-1}\partial\bar{\partial}\varphi$$

for some $\varphi \in C^\infty(M)$ which is unique up to addition of a constant.

14. Yau's Theorem

Theorem 14.1 (Version 1)**.** *Let (M,ω) be a compact Kähler manifold and $\Omega \in c_1(M)$ ($c_1(M) = \frac{1}{\pi}[\mathrm{Ric}(\omega)]$). Then there exist unique a Kähler metric $\tilde{\omega}$ in $[\omega]$ with*

$$\mathrm{Ric}(\tilde{\omega}) = \Omega.$$

In particular, if $c_1(M) = 0$, then there exist a Kähler metric $\tilde{\omega}$ in $[\omega]$ with $\mathrm{Ric}(\tilde{\omega}) = 0$.

Theorem 14.2 (Version 2)**.** *Let (M,ω) be a compact Kähler manifold. Given any volume form $e^F\omega^n$ on M with $\int_M e^F\omega^n = \int_M \omega^n$ for $F \in C^\infty(M)$. Then there exist a unique Kähler metric $\tilde{\omega} \in [\omega]$ with*

$$\tilde{\omega} = e^F\omega^n$$

is called Calabi-Yau equation.

We will show two versions is equivalent.

Proof. **Version 2$\Longrightarrow$ Version 1.**
Let $\Omega \in \pi c_1(M)$, then

$$\pi c_1(M) = [\mathrm{Ric}(\omega)] = [\Omega].$$

By $\partial\bar{\partial}$-Lemma, we have

$$\Omega - \mathrm{Ric}(\omega) = -\frac{\sqrt{-1}}{2}\partial\bar{\partial}F$$

for some $F \in C^\infty(M)$.
Using Version 2, i.e., $\tilde{\omega}^n = e^F \omega^n$, then

$$-\frac{\sqrt{-1}}{2}\partial\bar{\partial}\log\tilde{\omega}^n = -\frac{\sqrt{-1}}{2}\partial\bar{\partial}F - \frac{\sqrt{-1}}{2}\partial\bar{\partial}\log\omega^n = \Omega.$$

□

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